

DATA STRUCTURES AND ALGORITHMS DESIGN

SE ZG519/SS ZG519

Comprehensive Examination (Regular) Question Paper

#Questions: 4

Marks: 4 * 20 = 80

Q1 [Marks: 5 * 4 = 20]. Use master theorem to solve (show detailed steps) the following recurrence relations. In each of these cases, clearly specify the values of the parameters of ***a***, ***b***, ***k*** and ***p***, and finally show the solution in Big-O notation.

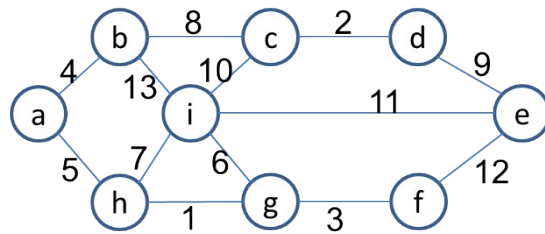
a) $T(n) = 2T\left(\frac{n}{2}\right) + \log n$

b) $T(n) = 16T\left(\frac{n}{2}\right) + (n \log n)^4$

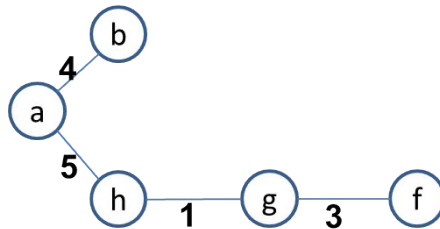
c) $T(n) = 7T\left(\frac{n}{3}\right) + n$

d) $T(n) = 9T\left(\frac{n}{3}\right) + n^3 \log n$

Q2 [Marks: (2+2) + (1+2) + (1 + 2 + 3 + 1) + (5+1) = 20]. Consider constructing minimal spanning tree (MST) for the given graph as shown below.



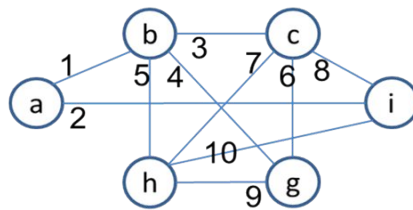
- a) Write down in plain English an example of NP-Hard problem for which an approximation algorithm can make use of MST of this graph. Explain your reason.
- b) Which data structure you can use to store this graph efficiently? Graphically show the representation in that data structure.
- c) Consider that the following tree has been generated by a MST constructing algorithm at an intermediate stage. Which between Prim's and Kruskal's algorithm has been used to construct the tree and why? Consider this tree as the starting point and apply Kruskal's algorithms to complete the MST construction. Show the final MST. What is the final cost of the MST?



- d) Following the MST in part c), now consider the problem of constructing second best MST for the given graph. Graphically show the second best MST. What is the cost of this MST?

Q3 [Marks: 3 + 3 + 3 + 6 + 3 + 2 = 20]. Consider the problem of vertex cover and answer the following questions.

- a) What is the minimum vertex cover of a complete graph with four nodes? Show your answer by drawing the graph.
- b) Write down the decision version of the problem of vertex cover.
- c) Write using pseudocode an approximation algorithm to compute minimum vertex cover of a graph.
- d) Use your algorithm in part c) to compute the minimum vertex cover of the following graph. Consider processing the edges according to the edge weights.



- e) What could be the optimal vertex cover of the graph in part c)?
- f) Explain the algorithm used in part e) in pseudocode.

Q4 [Marks: (1 +5) + (12+2) = 20]. Consider 0-1 knapsack problem.

a) In order to solve this problem using dynamic programming technique, assume that solution matrix be names as S, the maximum capacity of the knapsack is H, the function $\max()$ finds the maximum of two quantities, and an item and a weight value are indicated by t and h respectively. Write down the mathematical formula for solving the 0-1 knapsack problem along with the initial condition.

b) Using the formulation in part a), show the detailed solution steps when the maximum capacity $H = 10$, and the items are $(1,1)$, $(2,6)$, $(5,5)$, $(3,2)$ and $(4,20)$, where an item (x,y) indicates that the item has weight x and benefit y . To show your work, use table as shown below, the first row is filled for you. Write down the items selected.

[illegible]